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A

PROJECT

PROPOSAL ON

**AI-Driven Forecasting and Scenario Analysis of Labour
Market Imbalance in Canada**

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Introduction

Canada's labor market underwent significant changes between 2015 and 2025 as a result of economic cycles and the COVID-19 pandemic. During this time, a gap has become more visible between people looking for work and employers trying to fill jobs. Many workers remain unemployed, while employers report difficulty finding suitable candidates. This shows that workers and jobs are not matching well, which is referred to as labour market imbalance. Recent labour market analysis suggests that this mismatch has increased in Canada and has become an important concern for workforce and economic planning (Bank of Canada, 2025).

This project studies labour market imbalance in Canada using official data from Statistics Canada. The analysis compares the number of job vacancies with the number of unemployed people using the vacancy-to-unemployment ratio, which is commonly used in labour market studies (Statistics Canada, 2023). This measure is used to assess current labour market conditions, review historical trends, and forecast how labour imbalance may evolve over time. In addition, scenario-based analysis is applied to explore how potential changes in labour demand and labour supply could influence future imbalance. The overall aim of this study is to provide clear, data-driven insights that support workforce planning in Canada.

Problem Statement

Canada's labour market is facing a practical and ongoing problem. Employers across the country continue to report a large number of unfilled jobs, while at the same time many people remain unemployed. Current labour market analysis mainly explains recent conditions and does not provide enough guidance on what may happen next or how long this imbalance may continue (Statistics Canada, 2023).

This lack of forward-looking insight affects policymakers, employers, and workforce planners. When future labour market conditions are not anticipated, decisions are often made too late and only after problems appear. Statistics Canada has highlighted the need for better labour market information that supports forecasting and early planning rather than relying only on historical trends (Statistics Canada, 2024). This project addresses this problem by focusing on forecasting labour market imbalance and examining future scenarios, with the goal of supporting more timely and informed workforce planning in Canada.

Objectives

The objective of this study is to analyze labour market imbalance in Canada using a clear and data driven approach. The study focuses on understanding current conditions, identifying persistent mismatch, and estimating how imbalance may evolve in the future. The specific objectives are:

1. The first objective is to assess the current level of labour market imbalance in Canada by examining trends in employment, unemployment, and job vacancies.
2. The second objective is to analyze the interaction between labour demand and labour supply over time in order to explain why labour market imbalance persists.
3. The third objective is to forecast future labour market imbalance in Canada using time series forecasting model applied to job vacancy and unemployment data.
4. The final objective is to conduct scenario-based analysis to explore how potential changes in labour demand and labour supply may influence future labour imbalance.

Literature Review

Canadian labour market studies show that job vacancies and unemployment have moved in opposite directions in recent years, indicating persistent labour imbalance. Analysis from Statistics Canada shows that elevated job vacancies have coexisted with ongoing unemployment, particularly following the COVID-19 period, suggesting challenges in matching available workers with open positions (Statistics Canada, 2023). Bank of Canada further highlights that labour market tightness and worker–job mismatch have become more pronounced and pose risks to economic performance (Bank of Canada, 2025). Labour market imbalance is commonly examined using the Beveridge curve framework, which captures the relationship between job vacancies and unemployment and is widely used to assess matching efficiency and labour market tightness (Organisation for Economic Co-operation and Development, 2023). At the international level, the International Labour Organization emphasizes that persistent mismatch between labour demand and labour supply is a structural challenge affecting labour markets across advanced economies (International Labour Organization, 2024).

While existing studies provide strong descriptive evidence of labour imbalance, limited attention has been given to forecasting imbalance using vacancy-to-unemployment ratios or to exploring future conditions through scenario-based analysis. This study addresses this gap by extending the analysis toward forward-looking forecasting and scenario evaluation.

Research Questions

This study is guided by the following research questions:

1. What do trends in employment and job vacancies reveal about the current level of labour market imbalance in Canada?
2. How do labour demand and labour supply interact over time, and why does imbalance persist?
3. How is labour market imbalance in Canada expected to evolve based on historical data and AI-assisted forecasts?
4. Which scenario-based adjustments to labour demand or labour supply are most effective in reducing forecasted imbalance?

Hypotheses

Hypothesis 1: Persistence of Labour Market Imbalance

Canada's labour market is expected to exhibit persistent imbalance when high job vacancies coexist with ongoing unemployment over time.

H₀: Labour market imbalance does not persist in Canada over time.

H₁: Labour market imbalance persists in Canada, with elevated job vacancies and unemployment occurring simultaneously.

Rationale: In a well-functioning labour market, increases in job vacancies are typically followed by reductions in unemployment. If vacancies and unemployment remain high at the same time, it suggests that workers and jobs are not matching efficiently. This hypothesis will be examined by analyzing historical trends in the vacancy-to-unemployment ratio to determine whether imbalance persists rather than adjusting toward equilibrium.

Hypothesis 2: Labour Demand and Supply Interaction

Labour market imbalance is expected to persist when changes in labour demand are not matched by proportional changes in labour supply.

H₀: Changes in job vacancies are not related to changes in unemployment or labour force conditions.

H₁: Changes in job vacancies are related to changes in unemployment and labour force conditions, indicating labour market mismatch.

Rationale: When labour demand changes, unemployment and labour force conditions are expected to adjust accordingly. If these adjustments are weak or delayed, imbalance may continue. This hypothesis will be evaluated by examining the relationship between vacancies and unemployment over time to assess whether labour supply responds proportionately to changes in labour demand.

Hypothesis 3: Future Labour Market Imbalance

Based on recent labour market trends, imbalance in Canada is expected to continue rather than resolve quickly.

H₀: Labour market imbalance in Canada is expected to stabilize or decline in the future.

H₁: Labour market imbalance in Canada is expected to persist or increase in the future.

Rationale: If the structural patterns observed in recent years continue, labour market imbalance may remain elevated. This hypothesis will be assessed using Prophet-based forecasts of job vacancies and unemployment to evaluate projected trends in labour market imbalance over the forecast horizon.

Hypothesis 4: Scenario-Based Workforce Adjustments

Targeted changes to labour demand or labour supply are expected to influence future labour market imbalance.

H₀: Scenario-based adjustments to labour demand or labour supply do not reduce future labour market imbalance.

H₁: Scenario-based adjustments to labour demand or labour supply reduce future labour market imbalance.

Rationale: Labour market imbalance reflects how effectively workers are matched to available jobs. Adjustments to labour demand or labour supply may improve this alignment. This hypothesis will be tested by comparing baseline forecasts with scenario-adjusted forecasts to assess whether specific demand-side or supply-side changes reduce projected imbalance.

Methodology

This study will use a quantitative, data-driven approach to examine and forecast labour market imbalance in Canada. The analysis will begin with a review of historical labour market trends, followed by forecasting future imbalance and scenario-based evaluation to support workforce planning.

Data Sources

The study will use publicly available monthly labour market data from Statistics Canada. Labour demand will be measured using job vacancy data from the Job Vacancy and Wage Survey, while labour supply will be measured using employment, unemployment, and labour force data from the Labour Force Survey (Statistics Canada, 2024). The combined dataset will cover the period from mid-2015 to December 2024, providing approximately 110 monthly observations. All data will be national in scope and seasonally adjusted to ensure consistency over time.

Data Preparation and Indicator Definition

Monthly observations will be aligned across datasets and checked for missing values. Labour market imbalance will be measured using the vacancy-to-unemployment ratio, which compares the number of job vacancies with the number of unemployed persons. This ratio will serve as the main indicator throughout the study.

$$\theta_t = V_t / U_t$$

Where:

θ_t = Labour market imbalance at time t

V_t = Number of job vacancies at time t

U_t = Number of unemployed persons at time t

Analytical Approach

This study will adopt a structured analytical approach to examine labour market imbalance in Canada by combining descriptive, diagnostic, predictive, and scenario-based analysis.

The analysis will begin with descriptive analysis to review historical trends in employment, unemployment, and job vacancies. Labour market imbalance will be measured using the vacancy-

to-unemployment ratio, which provides a clear and widely used indicator of mismatch between labour demand and labour supply. Next, diagnostic analysis will be applied to examine how job vacancies and unemployment move relative to one another over time. This step will help explain why labour market imbalance persists rather than adjusting smoothly toward balance.

Predictive analysis will then be conducted using the Prophet time-series forecasting model to generate forward-looking estimates of job vacancies and unemployment. These forecasts will be used to assess how labour market imbalance may evolve in the future. Finally, scenario-based analysis will be applied to the forecast results to evaluate how potential changes in labour demand or labour supply may influence future labour market imbalance. Baseline forecasts will be compared with alternative scenarios to support forward-looking workforce planning.

Tools and Ethics

Python will be used for data preparation, analysis, and forecasting, including the implementation of the forecasting model. Excel will be used for data inspection and validation. Power BI or Tableau will be used to visualize trends, forecasts, and scenario results. GitHub will be used for version control, and Taiga will be used for project planning and task tracking.

The analysis will rely exclusively on publicly available and aggregated labour market data from Statistics Canada. No personal, confidential, or sensitive information will be used, and all data will be handled in accordance with academic and ethical research standards.

Expected Outcomes and Limitations

The study is expected to produce clear measures of labour market imbalance, forecasts of future imbalance, and scenario-based insights that support workforce planning decisions. The findings will help policymakers, labour market analysts, and employers better anticipate future labour market pressures. However, the results may be affected by data availability, unexpected economic shocks, or structural changes in the labour market. Forecast outcomes will depend on historical patterns and model assumptions, which will be acknowledged when interpreting results.

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